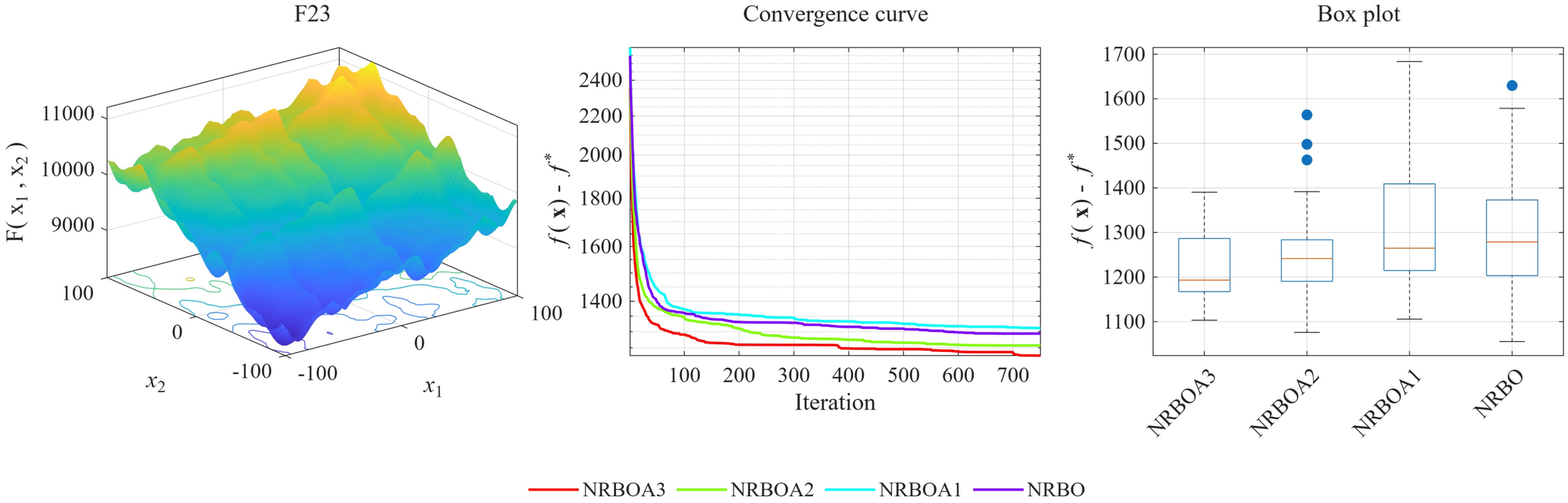

F23
F(x1 , x2)
11000
10000
9000
100
0
-100
x2
-100
0
x1
Convergence curve
f(x) - f*
2400
2200
2000
1800
1600
1400
100
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
1700
1600
1500
1400
1300
1200
1100
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO